A Non-Banach Class with Complete Collapse of Approximate Spectra

Run `fa_banach_001` — lane 3

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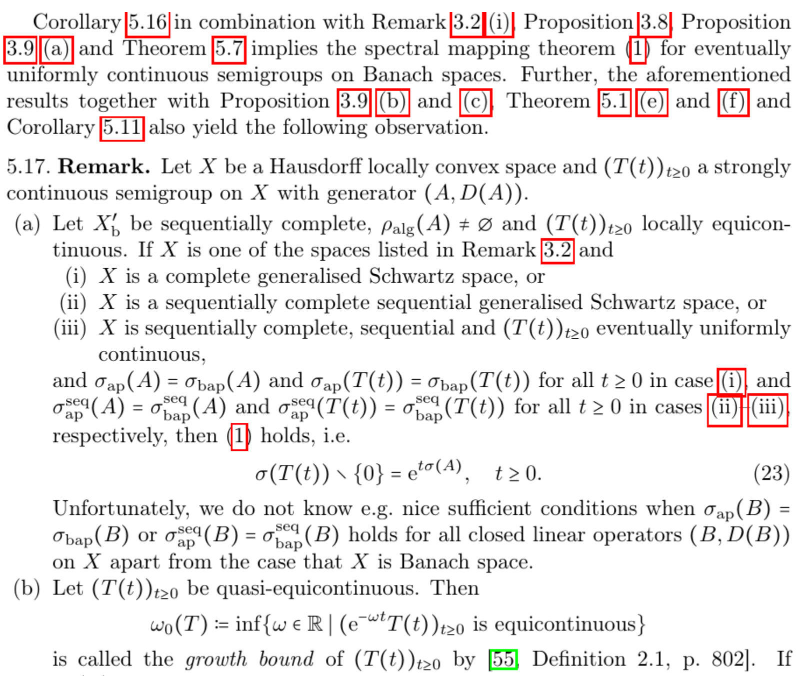
Status: claimed full answer, likely valid. If a complex vector space carries its strongest Hausdorff locally convex topology, then for every linear operator all four approximate point spectra in Remark 5.17(a) equal the point spectrum. The standard space $\varphi = \bigoplus_{n\geq 1}\mathbb{C}$ is complete, infinite-dimensional, and nonnormable, so this is genuinely beyond the Banach setting.

1 Source question

Karsten Kruse's *Spectral theory for semigroups on locally convex spaces*, arXiv:2509.11931v2, Remark 5.17(a), asks for useful sufficient conditions, apart from the Banach-space case, which ensure

$$\sigma_{\mathrm{ap}}(B) = \sigma_{\mathrm{bap}}(B) \qquad \text{or} \qquad \sigma_{\mathrm{ap}}^{\mathrm{seq}}(B) = \sigma_{\mathrm{bap}}^{\mathrm{seq}}(B)$$

for every closed linear operator $(B, D(B))$ on a Hausdorff locally convex space X [1].



Corollary 5.16 in combination with Remark 3.2 (i), Proposition 3.8, Proposition 3.9 (a) and Theorem 5.7 implies the spectral mapping theorem (1) for eventually uniformly continuous semigroups on Banach spaces. Further, the aforementioned results together with Proposition 3.9 (b) and (c), Theorem 5.1 (e) and (f) and Corollary 5.11 also yield the following observation.

5.17. **Remark.** Let X be a Hausdorff locally convex space and $(T(t))_{t\geq 0}$ a strongly continuous semigroup on X with generator $(A, D(A))$.

(a) Let X_b' be sequentially complete, $\rho_{\mathrm{alg}}(A) \neq \varnothing$ and $(T(t))_{t\geq 0}$ locally equicontinuous. If X is one of the spaces listed in Remark 3.2 and
  (i) X is a complete generalised Schwartz space, or
  (ii) X is a sequentially complete sequential generalised Schwartz space, or
  (iii) X is sequentially complete, sequential and $(T(t))_{t\geq 0}$ eventually uniformly continuous,

and $\sigma_{\mathrm{ap}}(A) = \sigma_{\mathrm{bap}}(A)$ and $\sigma_{\mathrm{ap}}(T(t)) = \sigma_{\mathrm{bap}}(T(t))$ for all $t \geq 0$ in case (i), and $\sigma_{\mathrm{ap}}^{\mathrm{seq}}(A) = \sigma_{\mathrm{bap}}^{\mathrm{seq}}(A)$ and $\sigma_{\mathrm{ap}}^{\mathrm{seq}}(T(t)) = \sigma_{\mathrm{bap}}^{\mathrm{seq}}(T(t))$ for all $t \geq 0$ in cases (ii)–(iii), respectively, then (1) holds, i.e.

$$\sigma(T(t)) \setminus \{0\} = \mathrm{e}^{t\sigma(A)}, \quad t \geq 0. \tag{23}$$

Unfortunately, we do not know e.g. nice sufficient conditions when $\sigma_{\mathrm{ap}}(B) = \sigma_{\mathrm{bap}}(B)$ or $\sigma_{\mathrm{ap}}^{\mathrm{seq}}(B) = \sigma_{\mathrm{bap}}^{\mathrm{seq}}(B)$ holds for all closed linear operators $(B, D(B))$ on X apart from the case that X is Banach space.

(b) Let $(T(t))_{t\geq 0}$ be quasi-equicontinuous. Then

$$\omega_0(T) := \inf\{\omega \in \mathbb{R} \mid (\mathrm{e}^{-\omega t}T(t))_{t\geq 0} \text{ is equicontinuous}\}$$

is called the *growth bound* of $(T(t))_{t\geq 0}$ by [55, Definition 2.1, p. 802]. If

Figure 1: Remark 5.17(a) in arXiv:2509.11931v2, PDF page 34.

2 Definitions and answer

The *strongest Hausdorff locally convex topology* on a complex vector space X is the topology generated by all seminorms on X. Equivalently, every linear map from X into any locally convex space is continuous.

For a linear operator $(B, D(B))$ and $\lambda \in \mathbb{C}$, the source defines $\lambda \in \sigma_{\mathrm{ap}}(B)$ when there is a net $(x_i) \subseteq D(B)$ which does not converge to zero but satisfies $(B - \lambda)x_i \to 0$. The symbol σ_{bap} requires the witnessing net to be bounded, and the superscript seq restricts to sequences. The point spectrum $\sigma_{\mathrm{p}}(B)$ consists of the eigenvalues.

3 Proof intuition

An approximate eigenvalue outside the point spectrum is exactly a failure of continuity of the algebraic inverse

$$(\lambda - B)^{-1} \colon \operatorname{ran}(\lambda - B) \longrightarrow X.$$

On a space with the strongest locally convex topology, every subspace again has the strongest locally convex topology. Therefore every linear map out of the range—in particular this inverse—is automatically continuous. No nonzero approximate eigen-net can exist unless a genuine eigenvector already exists. This argument does not need boundedness, sequentiality, or closedness of the operator.

4 The full collapse theorem

Lemma 1 (heredity of the strongest topology)**.** *Let X carry its strongest Hausdorff locally convex topology and let $Y \subseteq X$ be a linear subspace. Then the topology induced on Y is the strongest Hausdorff locally convex topology on Y. Consequently, every linear map from Y to a Hausdorff locally convex space is continuous.*

Proof. Let q be any seminorm on Y. Choose an algebraic complement Z with $X = Y \oplus Z$, and let $P \colon X \to Y$ be the associated algebraic projection. Then $q \circ P$ is a seminorm on X, hence is continuous by the definition of the topology of X. Its restriction to Y is q, so every seminorm on Y is continuous for the induced topology. This is precisely the strongest locally convex topology on Y.

If $L \colon Y \to F$ is linear and p is a continuous seminorm on F, then $p \circ L$ is a seminorm on Y and is therefore continuous. Hence L is continuous. □

Theorem 1. *Let X be a complex vector space with its strongest Hausdorff locally convex topology. For every linear operator $(B, D(B))$ on X—not necessarily closed or densely defined—one has*

$$\boxed{\sigma_{\mathrm{p}}(B) = \sigma_{\mathrm{ap}}(B) = \sigma_{\mathrm{ap}}^{\mathrm{seq}}(B) = \sigma_{\mathrm{bap}}(B) = \sigma_{\mathrm{bap}}^{\mathrm{seq}}(B).}$$

In particular, both equalities requested in Remark 5.17(a) hold for every closed operator on X.

Proof. Fix $\lambda \notin \sigma_{\mathrm{p}}(B)$. Then $T := \lambda - B \colon D(B) \to X$ is injective. Put $R := \operatorname{ran} T \subseteq X$. The algebraic inverse

$$S := T^{-1} \colon R \longrightarrow X$$

is linear. By Lemma 1, R has the strongest locally convex topology, and hence S is continuous.

If a net $(x_i) \subseteq D(B)$ satisfies $(B-\lambda)x_i \to 0$, then

$$x_i = S(Tx_i) = -S((B-\lambda)x_i) \longrightarrow 0.$$

Thus $\lambda \notin \sigma_{\mathrm{ap}}(B)$, and therefore it belongs to none of the three smaller approximate spectra. This proves that all four approximate point spectra are contained in $\sigma_{\mathrm{p}}(B)$.

Conversely, if $Bx = \lambda x$ for some $x \neq 0$, the constant net and constant sequence with value x are bounded, do not tend to zero, and have zero residual. Hence λ belongs to all four approximate point spectra. □

Corollary 1 (a concrete complete non-Banach class)**.** *Let*

$$\varphi = \mathbb{C}^{(\mathbb{N})} = \bigoplus_{n\geq 1} \mathbb{C}$$

carry its locally convex direct-sum topology, equivalently its strongest locally convex topology. Then φ is complete, infinite-dimensional, and nonnormable, and the five spectra in Theorem 1 coincide for every linear operator on φ.

Proof. The spectral assertion is Theorem 1. The space φ is the strict locally convex inductive limit of the finite-dimensional spaces $\mathbb{C}^n$. It is the standard complete Montel (DF)-space with the strongest locally convex topology; see, for example, Kąkol–Saxon [2]. It is nonnormable: an infinite-dimensional normed space admits discontinuous algebraic linear functionals, whereas every linear functional on φ is continuous by definition of its topology. □

Thus the sufficient condition is compatible with completeness and is not a disguised Banach assumption. The same proof works in arbitrary Hamel dimension for the strongest locally convex topology; φ is singled out because it is a standard sequential complete nonnormable example relevant to the source's semigroup framework.

5 Sharp contrast with Montel and nuclear hypotheses

The strongest-topology hypothesis cannot be replaced merely by familiar compactness or regularity assumptions. Let

$$s = \left\{ x = (x_n) : p_k(x) := \sup_{n\geq 1} n^k |x_n| < \infty \text{ for every } k \in \mathbb{N}_0 \right\}$$

be the nuclear Fréchet–Montel space of rapidly decreasing sequences, and define $(Bx)_n = e^{-n}x_n$. Then B is continuous, closed, and injective, but

$$0 \in \sigma_{\mathrm{ap}}^{\mathrm{seq}}(B) \setminus \sigma_{\mathrm{bap}}(B).$$

Indeed, the unit vectors e_m do not tend to zero since $p_0(e_m) = 1$, while $p_k(Be_m) = m^k e^{-m} \to 0$ for every k. Conversely, if (x_i) is bounded and $Bx_i \to 0$, then for fixed k and $C = \sup_i p_{k+1}(x_i)$,

$$\sup_{n>N} n^k |x_{i,n}| \leq C/N,$$

whereas the finitely many coordinates $n \leq N$ tend to zero. Hence $p_k(x_i) \to 0$ for every k, so $x_i \to 0$. This rules out completeness, metrizability, nuclearity, Fréchet, Schwartz, or Montel hypotheses—even in combination—as sufficient conditions by themselves.

6 Scope, verification, and novelty check

Theorem 1 fully answers the source's request for a useful sufficient condition beyond Banach spaces; it does not claim to characterize all locally convex spaces having the property. The theorem is stronger than requested because it treats arbitrary operators and both net and sequential versions simultaneously.

The proof was checked directly against the definitions in the current June 2026 arXiv v2. Its only structural step, Lemma 1, is proved from an algebraic complement and the defining family of all seminorms. A bounded novelty search through 19 July 2026 covered the run registry, the local arXiv source corpus, the current source paper and its references, arXiv, and exact web phrases combining "approximate point spectrum" with "strongest/finest locally convex topology" and φ. It found the standard topology literature and unrelated approximate-spectrum literature, but no prior statement of Theorem 1 or its application to Remark 5.17(a). The topology facts are classical; novelty of the spectral observation remains subject to expert review.

References

[1] K. Kruse, *Spectral theory for semigroups on locally convex spaces*, J. Math. Anal. Appl. **564** (2026), Article 130889; arXiv:2509.11931v2.

[2] J. Kąkol and S. A. Saxon, *Montel (DF)-spaces, sequential (LM)-spaces and the strongest locally convex topology*, J. London Math. Soc. (2) **66** (2002), 388–406, doi:10.1112/S0024610702003459.